Scheme of valuation

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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECSC201 - INTRODUCTION TO MOBILE AND WEB TECHNOLOGIES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A Very Short Answer Questions Answer all questions Each question carries 2 marks

1. **Define the term World Wide Web. [K] / [CO1]**
   The World Wide Web (WWW) is a collection of interlinked web pages accessible over the Internet using browsers. [2 marks]
2. **What is a flexible grid layout? [K] / [CO2]**
   A flexible grid layout uses relative units like percentages instead of fixed pixels to create responsive web pages that automatically adjust to different screen sizes.
3. **Define client-server model. [K] / [CO2]**
   The client-server model is a network architecture where the client requests services and the server provide them. (define client and server).
4. **Name two popular mobile development frameworks and briefly explain one key feature of each. [K] / [CO3].**
   React Native, Flutter, and Xamarin [List any two – 1 mark; list one feature of each – 1 mark]
5. **If a business wants a single codebase that works across devices, why choose a PWA? [U] / [CO4**]
   A Progressive Web App (PWA) works across devices using a single codebase and provides offline support, push notifications, and app-like experiences without installation. [list any two advantages]

Part B
Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

6. **Discuss the role of AI in today's mobile/web applications. [U] / [CO1]**

Write any five roles of AI in today's Mobile/Web Applications give 5 marks

a) Personalization of content – recommends products or media based on user behavior.
b) Chatbots and virtual assistants – provide instant customer support and voice interaction.
c) Predictive analytics – forecasts user needs and preferences using data patterns.
d) Voice and image recognition – enables speech commands, face unlock, and visual search.
e) Enhanced security – detects fraud and unusual activities using pattern recognition.
f) Automation – simplifies repetitive tasks and improves efficiency.
g) Smart decision-making – supports data-driven insights for businesses.
h) Improved user experience (UX) – makes apps more responsive and adaptive to user needs.

7. **What is the process for running Node.js code, and what is one key requirement for its execution? [U] / [CO2]**

a. Node.js – Definition [2 mark]: Node.js is an open-source, cross-platform, runtime environment that allows developers to run JavaScript code outside of a web browser, typically on the server side.
3 marks for below points.
b. Install Node.js on your system from the official website (https://nodejs.org).
c. Create a JavaScript file (e.g., app.js) containing the Node.js code.
d. Open Command Prompt / Terminal and navigate to the file location.
e. Run the file using the command:
node app.js

8. **What is React, and how does its component-based architecture and virtual DOM contribute to building modern web applications? Provide an example of a React component in use [U] / [CO2]**

a. Definition - React [1 mark]: React is an open-source JavaScript library developed by Meta (Facebook) for building user interfaces (UIs) — especially for single-page and dynamic web applications.

b. Component-Based Architecture [1.5 marks]**:**
   - UI is divided into reusable components.
   - Each component manages its own state and logic.
   - Makes development faster and maintenance easier.
     Example: A Header, Footer, and Login Form can each be independent components.

c. Virtual DOM [1.5 marks]:
   - React uses a Virtual DOM (a lightweight copy of the real DOM).
   - When changes occur, react updates only the changed parts instead of reloading the entire page.
   - This improves performance and speed.

d. Example of a React component in use [1 mark]
   Functional component or class component [any one]

9. Explain the key stages of a mobile application's lifecycle [A] / [CO3]

Ideation and conceptualization, Planning and Analysis, Design, Development (frontend, backend), testing, deployment, maintenance. [Architecture: 1 mark, explain each stage 4 marks].

10. Explain the importance of user experience (UX) in mobile applications and how it affects app success. [U] / [CO3]

[List any 5 importance give 5 marks]

a) Defines User Satisfaction:
   - A good UX ensures the app is easy, enjoyable, and efficient to use.
   - Satisfied users are more likely to keep using the app.

b) Improves User Retention:
   - Users stay longer and return frequently when the app feels intuitive and responsive.

c) Increases App Ratings and Reviews:
   - Positive UX leads to better feedback on app stores, attracting more downloads.

d) Enhances Brand Image:
   - A smooth, well-designed app builds trust and strengthens the brand's reputation.

e) Boosts Conversions and Business Growth:
   - Good UX helps users complete goals easily (e.g., purchases, sign-ups), increasing revenue.

f) Reduces Uninstalls:

- Poor navigation, slow performance, or confusing layouts lead users to uninstall the app quickly.

11. Illustrate AES encryption and its use in mobile security. [U] / [CO4]

a) Definition [1 mark]:
   - AES (Advanced Encryption Standard) is a symmetric key encryption algorithm used to secure sensitive data.
   - The same key is used for both encryption and decryption.
   - AES supports 128-bit, 192-bit, and 256-bit key lengths.
   - Larger key sizes provide stronger security.

b) Working Principle [1 mark] and figure [2 marks]:
   - Data is divided into blocks of 128 bits.
   - Each block passes through multiple rounds of substitution, permutation, and mixing to convert plaintext into ciphertext.

c) Use in Mobile Security [1 mark]:
   - Protects stored data such as passwords, tokens, and files.
   - Secures data transmission between mobile apps and servers.
   - Used in Android Keystore and iOS Keychain for safe key storage.

12. A developer needs to store authentication tokens securely on Android. Which secure storage mechanism should be used? [U] / [CO4]

Explain Android Keystore System [2 mark]
EncryptedSharedPreferences- [2 marks]
Benefit [1 mark]

13. List two security best practices for web application development, and briefly explain how each protects the application. [U] / [CO4]

[Explain any two give 5 marks, (2.5 marks for each practice)]

a) Use HTTPS (SSL/TLS) for secure data transmission.
b) Validate all user inputs to prevent SQL injection and XSS attacks.
c) Implement strong authentication (e.g., OAuth, 2FA).
d) Use parameterized queries in database operations.
e) Hash and salt passwords before storing them.
f) Keep software and dependencies updated.
g) Use secure session management with timeouts.
h) Limit user permissions and follow least privilege principle.
i) Implement Content Security Policy (CSP).

j) Regularly perform security audits and penetration testing.

Part C- Essay Questions

Answer any 2 questions

Each question carries 15 marks

14. How do we create frames? Explain with suitable example. [U] / [CO2]

- Definition (Concept)- 2 marks
- Purpose / Use – 2 marks
- Tags & Syntax – 4 marks
- Example – 7 marks

15. Provide a detailed analysis of the different types of mobile applications — native, hybrid, and web apps. Discuss their underlying technologies, development strategies, advantages, and disadvantages. Justify your selection for developing an e-commerce application for a startup [E] / [CO3]

- Native Application: definition, advantages, disadvantages, technologies [4 marks]
- Hybrid application: definition, advantages, disadvantages, technologies [4 marks]
- Web Application: definition, advantages, disadvantages, technologies [4 marks]
- Justification for E-Commerce Startup – [3 marks]

16. Explain how AI-driven personalization improves user experience in mobile and web applications. [U] / [CO4]

- Key Techniques Used [3 marks]
- Ways AI Improves User Experience [any 5 with explanation give 10 marks]

a) Analyzes user behavior – AI tracks browsing history, clicks, and preferences.
b) Provides personalized recommendations – Suggests products, movies, or content (e.g., Amazon, Netflix).
c) Uses predictive analytics – Anticipates user needs and offers relevant suggestions.
d) Improves search accuracy – Delivers smarter, faster search results.
e) Enables adaptive UI – Changes layout or features based on user habits.
f) Offers context-aware notifications – Sends timely, relevant alerts and offers.

g) Enhances customer support – AI chatbots provide instant, personalized assistance.
h) Boosts engagement – Personalized experiences keep users active and loyal.
i) Saves user time – Reduces effort by predicting actions and preferences.
j) Improves satisfaction and retention – Users feel valued and understood.

- Practical Examples [2 marks]

References:

1. "Mobile Application Development: A Practical Approach" by Jeff McWherter and Scott Gowell
2. "Web Security for Developers: Real Threats, Practical Defense" by Malcolm McDonald
3. "HTML and CSS: Design and Build Websites" by Jon Duckett
4. "JavaScript and JQuery: Interactive Front-End Web Development" by Jon Duckett
5. "Responsive Web Design with HTML5 and CSS" by Ben Frain